ANALYZING SEVERITY OF US CAR ACCIDENTS (2016-2023)

Uncovering how weather, location, and infrastructure are associated with traffic delays

WHO WE ARE

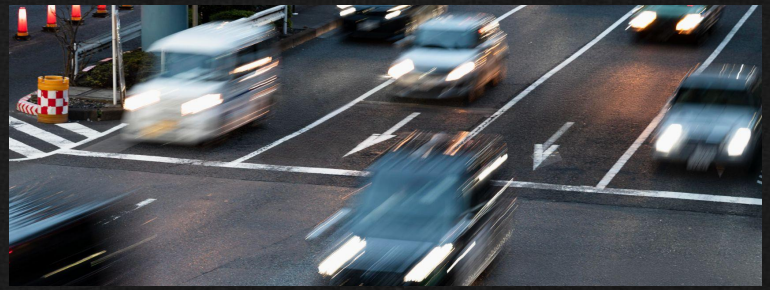
We are an Emergency Services Data Analytics Team overseeing car accidents throughout the United States.

Our goal is to **reduce crash severity** by analyzing what factors contribute to crashes and how we can make the roads safer for everyone!



RESEARCH QUESTION AND OBJECTIVE





Research Proposal

What factors relate to severity of an accident?

Objectives

- Analyze what factors contribute to crashes
- Examine what factors decrease severity
- Indicate what we need to implement to keep roads safer

DATA OVERVIEW AND PREPROCESSING
From Raw Data to Analysis-Ready Data

Dataset Overview

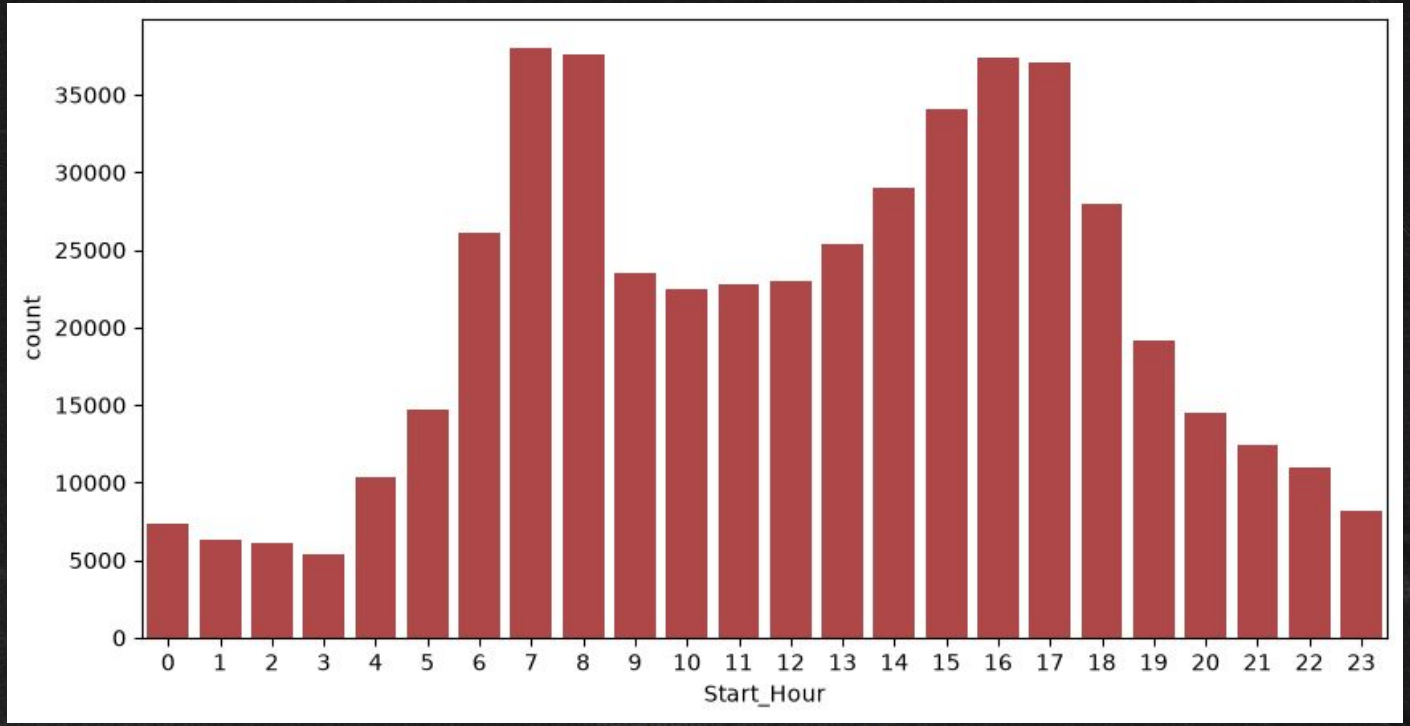
- **Source:** US Accidents (2016-2023)
- **Original dataset:** 500,000 observations and 46 variables.
- **Final dataset:** 500,000 observations and 31 variables.
- **Four analysis categories:** Time, Weather, Geography, and Road Infrastructure

Data Preprocessing

- Standardized date and time formats for time variables.
- Cleaned numerical weather measurements and grouped weather conditions into broader categories.
- Selected relevant geographic and roadway infrastructure features.
- Merged four cleaned datasets using accident IDs and verified row alignment.
- Preserved missing observations rather than automatically replacing them with zero.

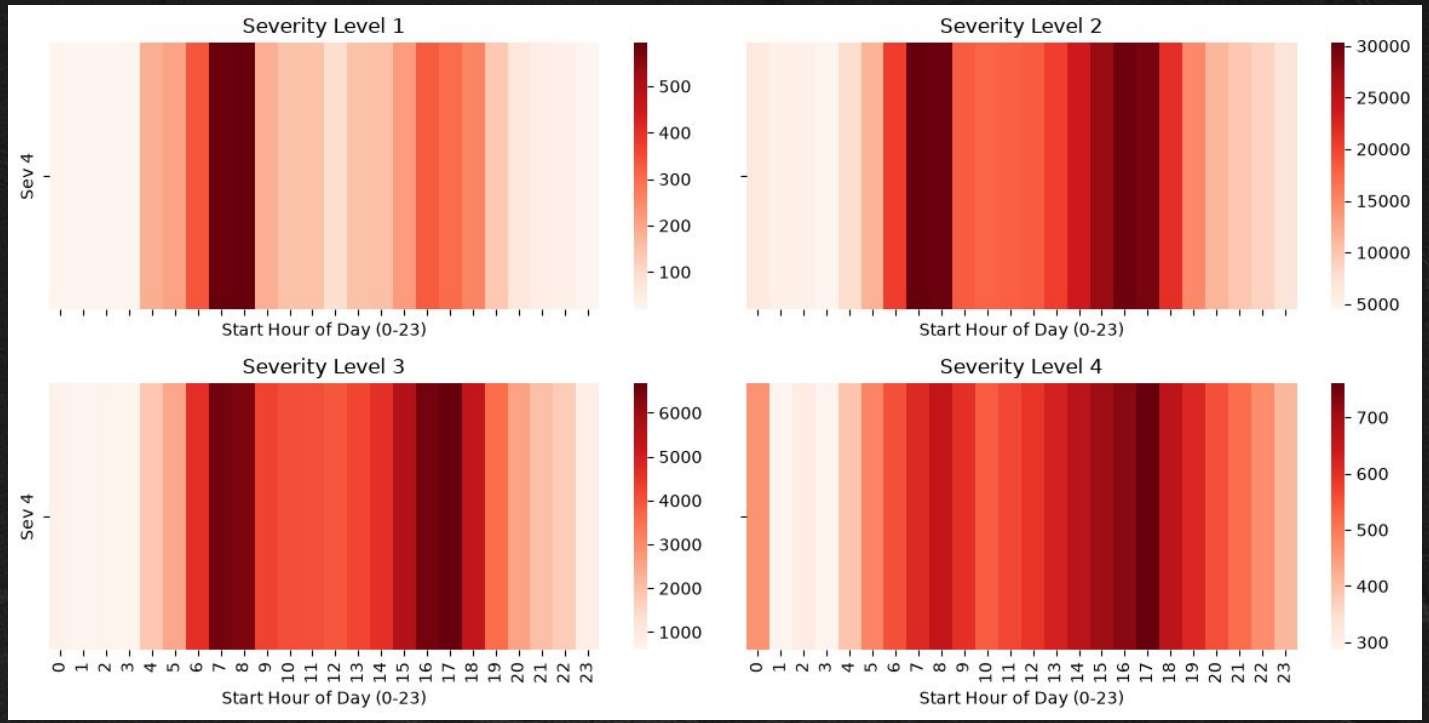
TIME FACTORS

The data follows a saddle shape. We can see peaks of accidents happen at rush hour 8AM and 4PM



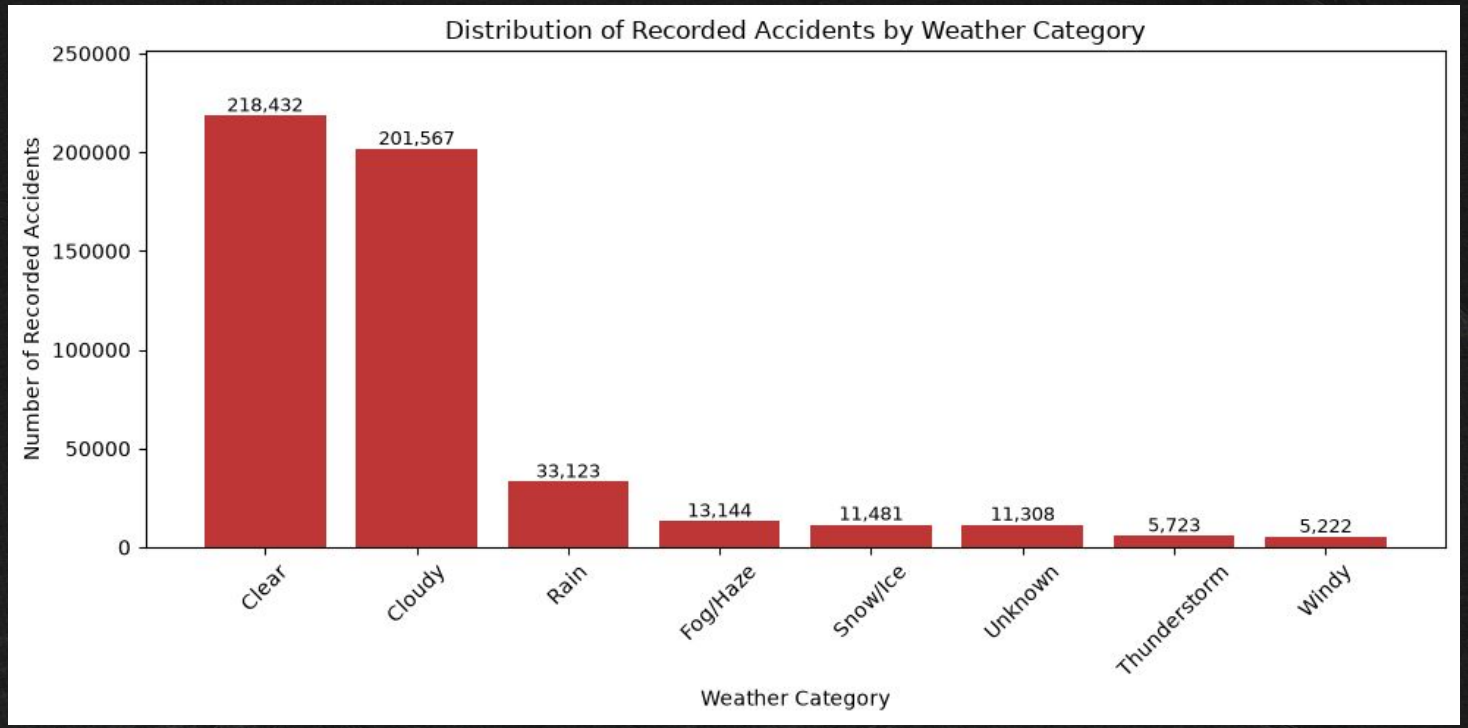
Time and Severity

What is the distribution of crashes per hour separated by their severity level



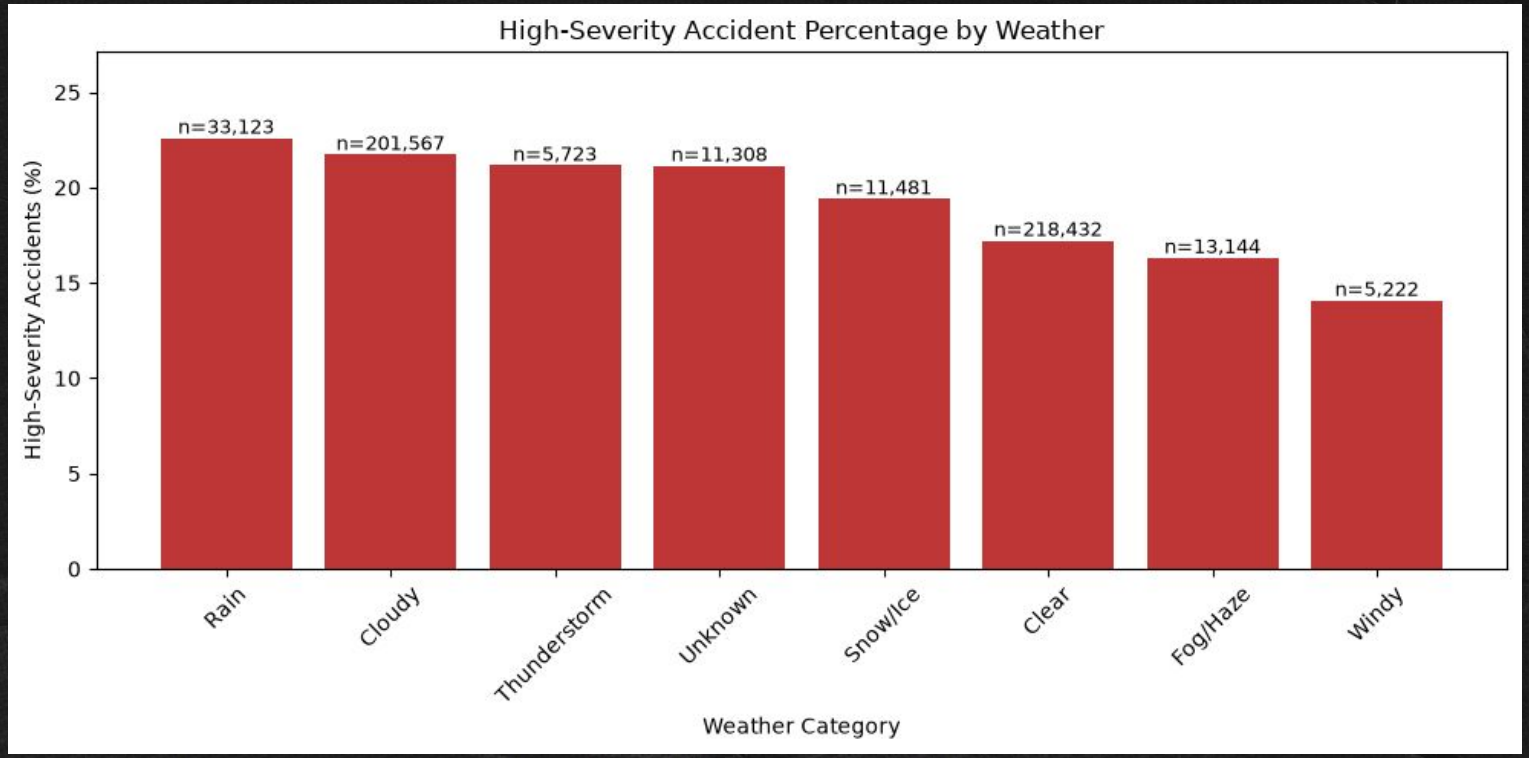
WEATHER FACTORS

What is the recorded number of accidents per weather category?



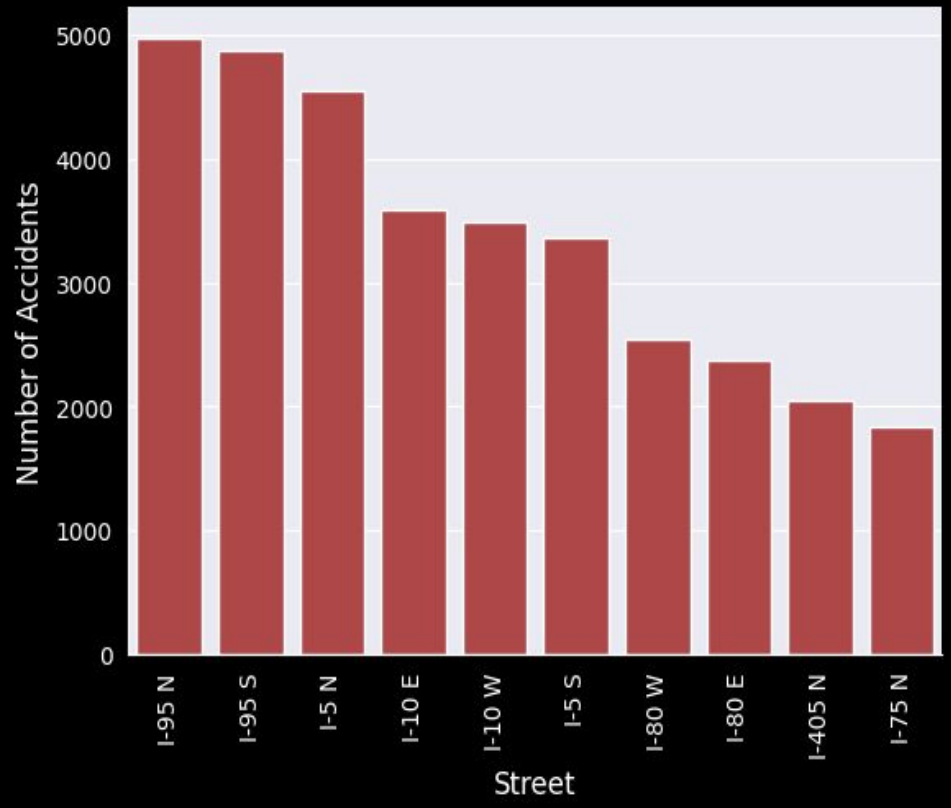
WEATHER FACTORS

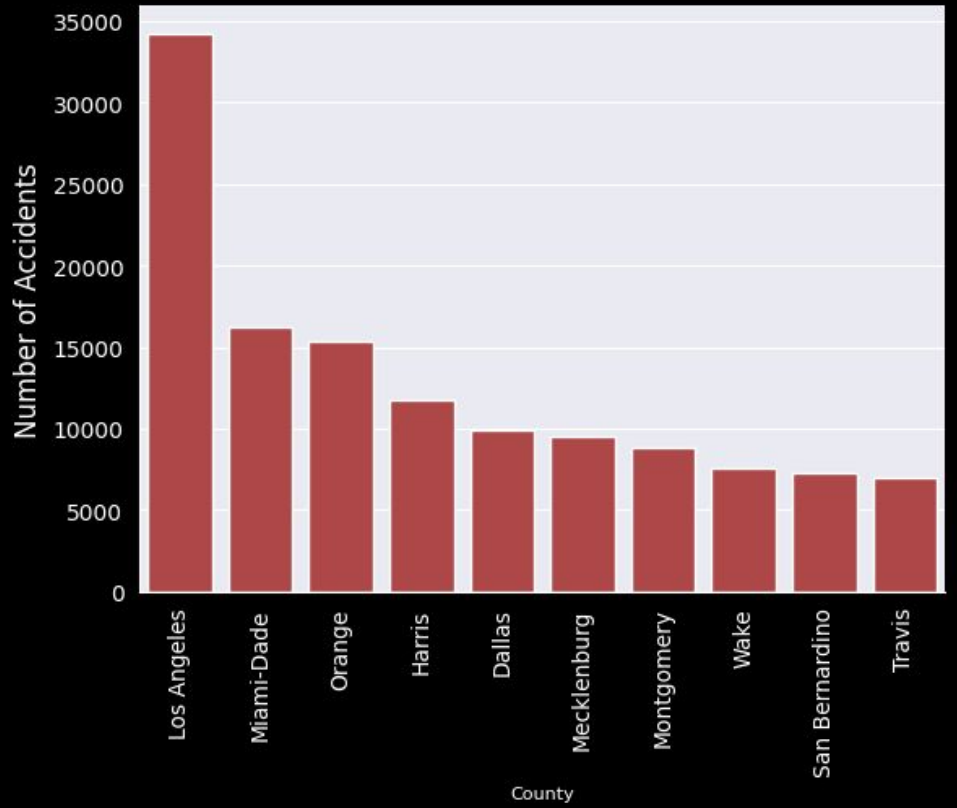
How does the proportion of high-severity accidents vary by weather conditions?



GEOGRAPHIC FACTORS

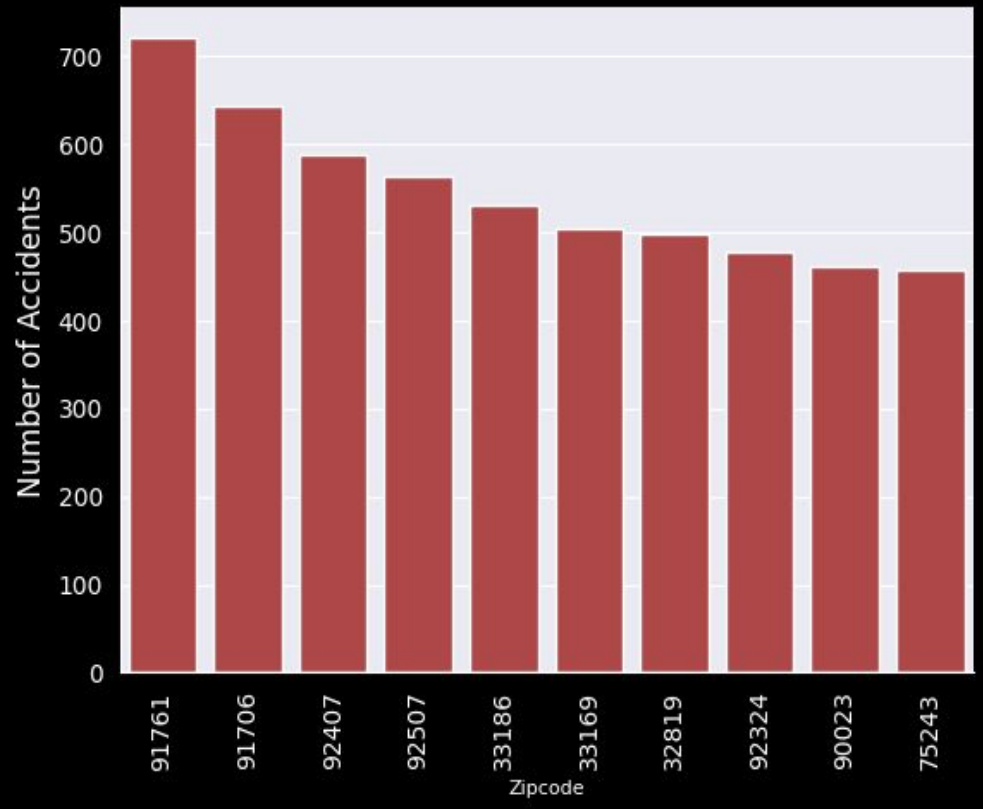
What regions have the most frequent number of accidents?

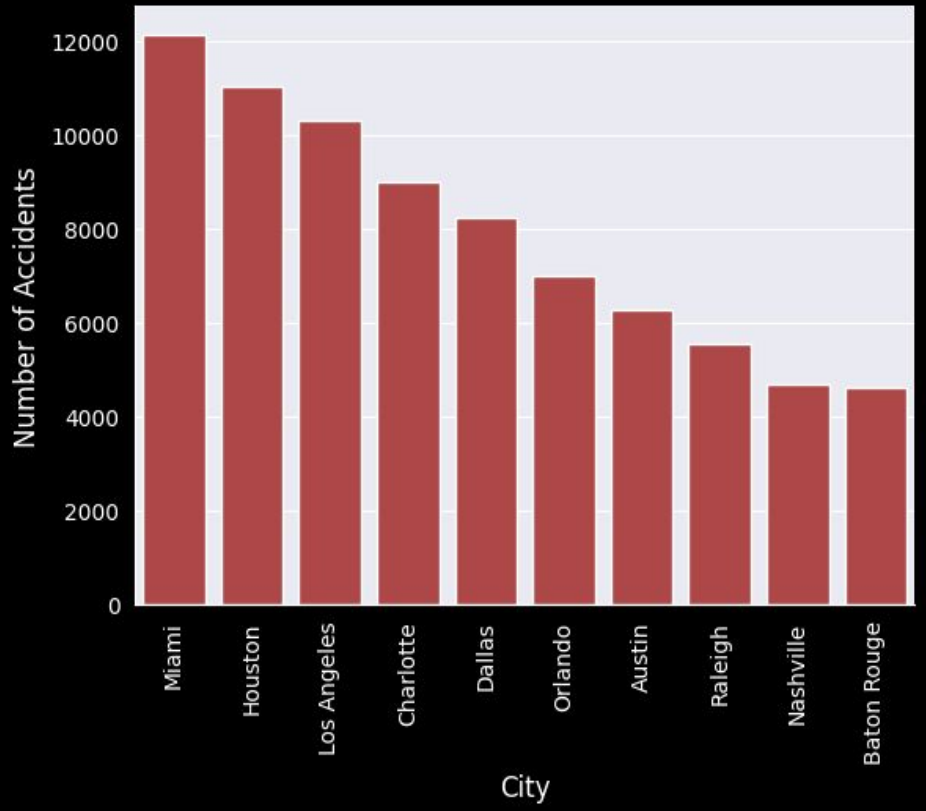




GEOGRAPHIC FACTORS

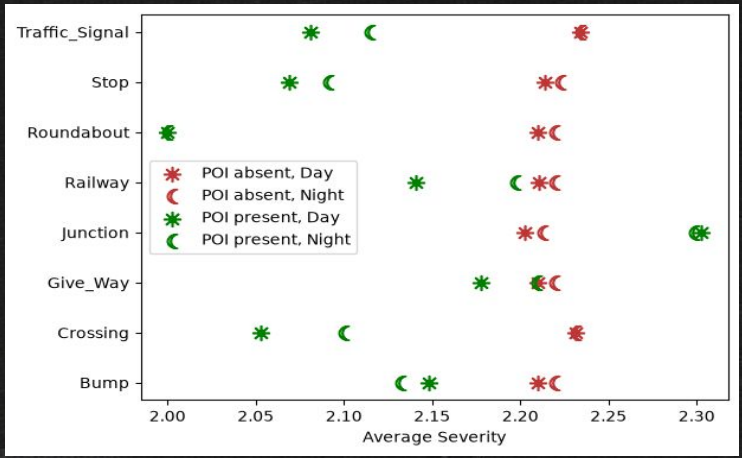
What regions in the US have the most frequent accidents?





ROAD INFRASTRUCTURE FACTORS

What road features typically correlate with the most frequent/severe accidents?

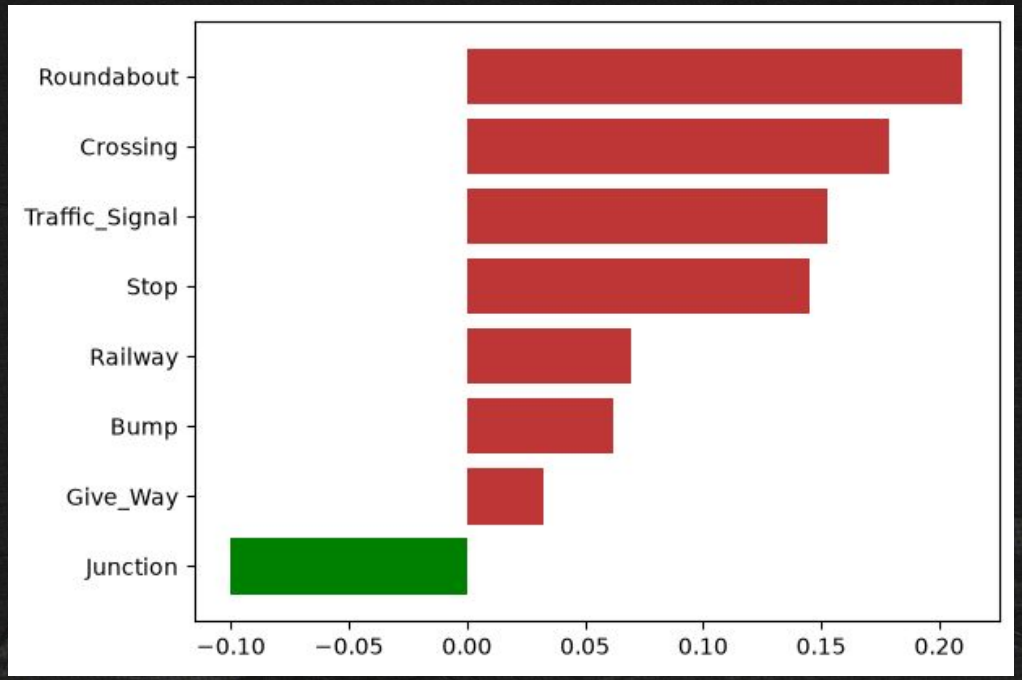


- In all cases but 1, the **<u>presence</u>** of a POI was associated with lower average recorded severity of accidents
  - The exception - Junction

- Nighttime in most cases does not have a great impact on crash severity
  - Night has a slightly higher severity factor on railways, crossings, and give ways

ROAD INFRASTRUCTURE FACTORS

Depending on budget: These are the areas to focus on, with the top having higher precedence



Limitations Regarding Unused Data Points

1. **Unused variables:** The original US Accidents dataset contained 46 variables, while our analysis focused on 31 variables, excluding other potentially relevant variables
2. **Missing observations:** Some variables were retained but have missing values, so not every accident can be included in every single analysis.
3. **Missing external context:** The dataset contains recorded accidents, but not necessarily the traffic exposure, driver behavior, phone usage, or road conditions needed to explain why accidents happened.

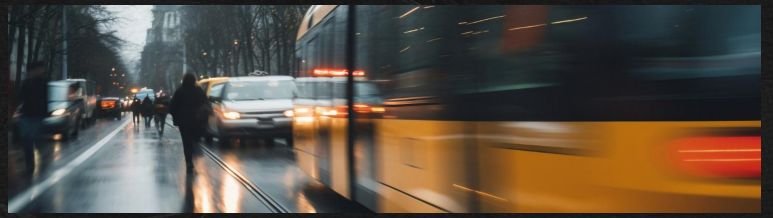
Limitations on Current Data Points

1. **Missing Values:** Precipitation (28.5%), accident distance (9.2%), and wind speed (7.4%) contain incomplete/missing observations.
2. **Severity Imbalance:** 79.6% of recorded accidents are classified as Severity 2, limiting representation of other severity levels.
3. **Weather Categories:** Clear and Cloudy conditions dominate the data, while severe weather categories contain substantially fewer observations.
4. **Measurement Limitations:** About 42.7% of accidents have a recorded distance of zero, limiting interpretation of roadway impact.
5. **Distance Values:** Around 10% of distance values were discovered to be potential outliers

Recommendations to Make Roads Safer

Recommendations for Future:

1. Provide better alternative travel during peak rush hour
    a. Offering better public transit can reduce the amount of cars on the road and help reduce the severity of accidents
2. SMS Dangerous Weather Alert App
    a. Similar to Amber Alerts a weather alert can be sent out to drivers when dangerous driving weather is detected
3. Add more road infrastructure
    a. Improve roadway drainage to help with rain caused accidents
    b. Add roundabouts, crossings, traffic signals to help decrease severity of accidents







THANK YOU!

Any questions?

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